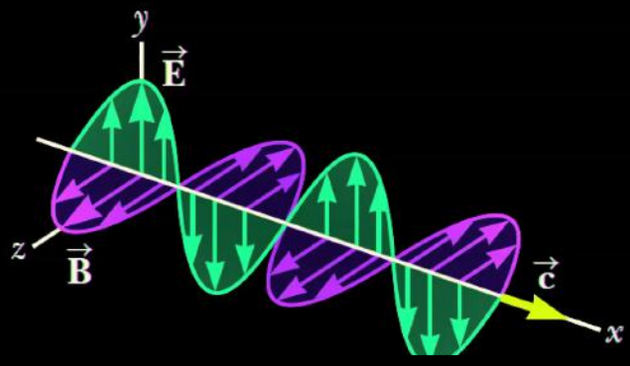
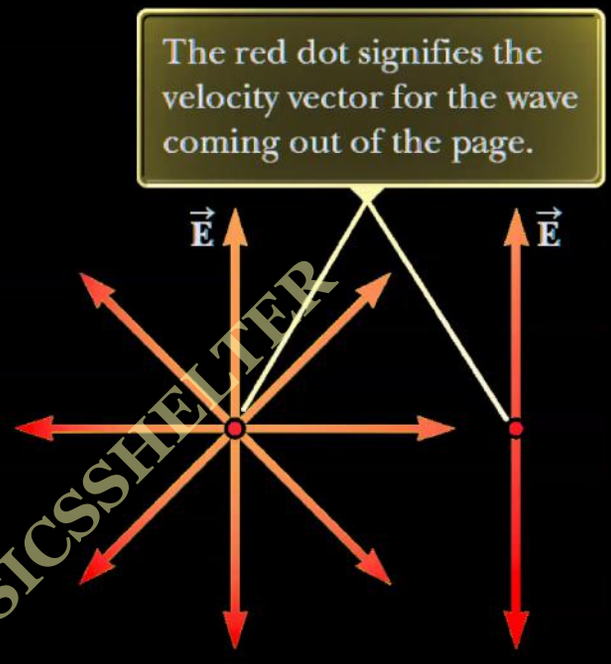
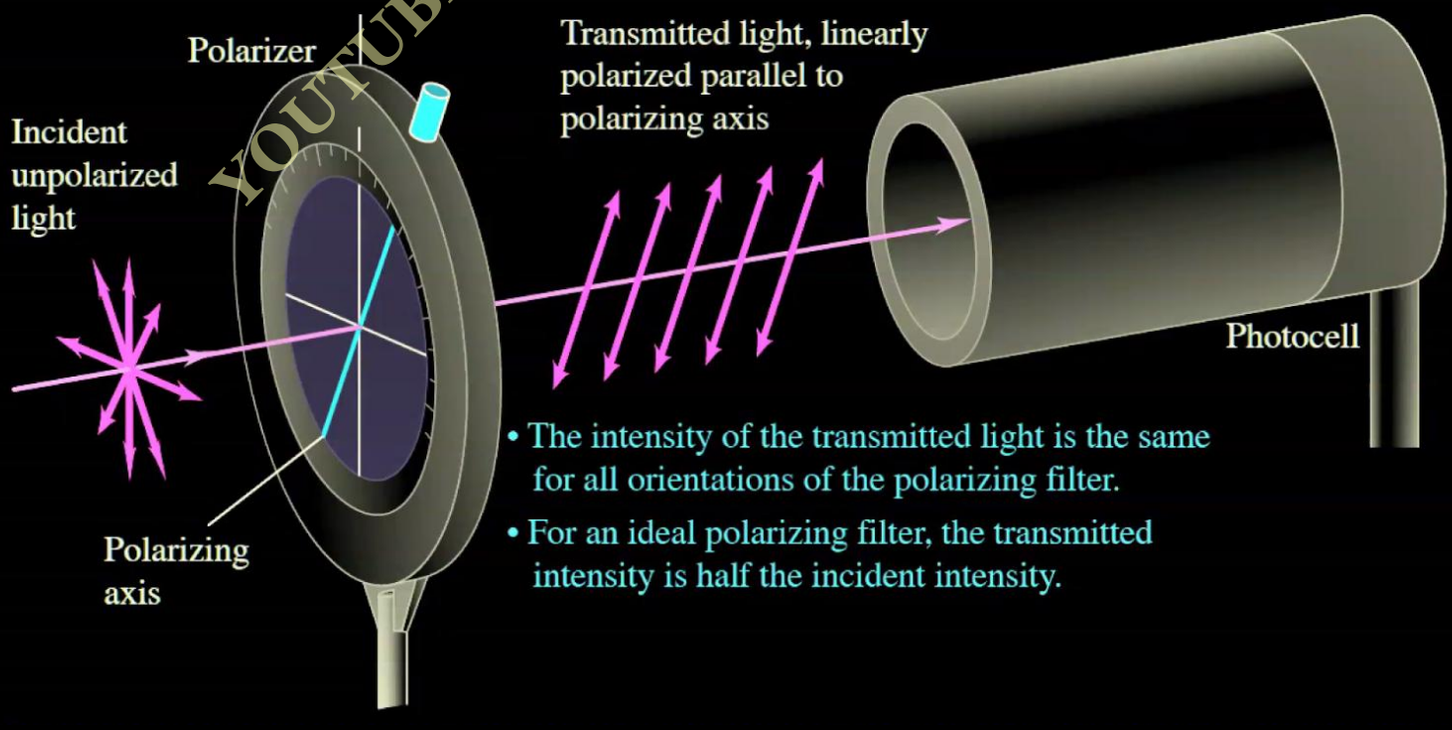
UNPOLARIZED LIGHT:

An ordinary beam of light from a light source consists of a large number of waves emitted by atoms of the light source. Recall that electric field vector is ⊥ to the direction of propagation of light. The direction of polarization of each wave is defined as the direction in which electric field is vibrating. However since waves generated from different atoms in a particular direction may not produce electric field oscillations in same direction, electric field at a given point may be pointing anywhere in a 360° angle in plane ⊥ to direction of propagation and this direction can vary randomly from time to time. When direction of oscillations in a beam of light are randomly directed with time we say that the light is unpolarized.





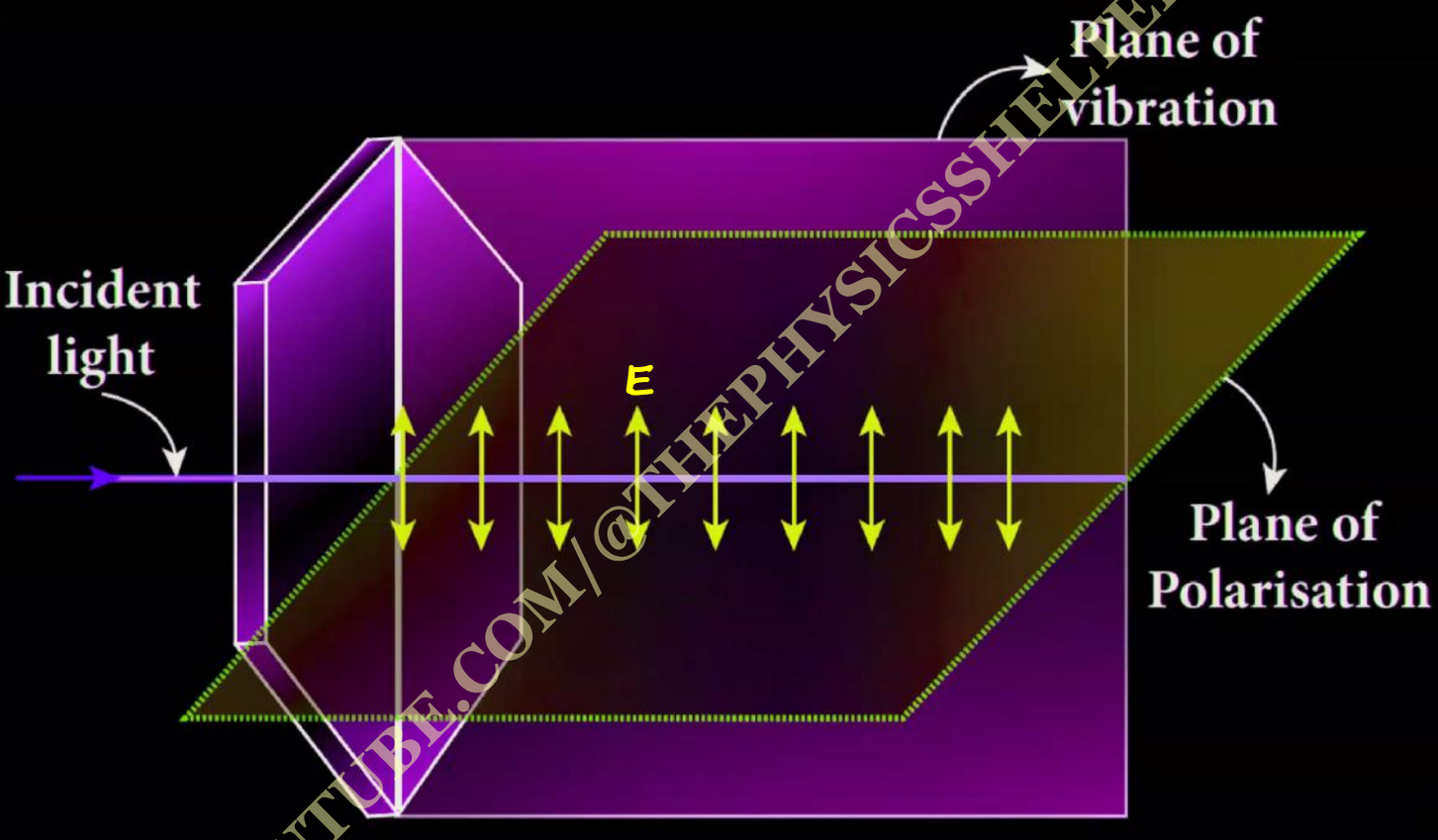
PLANE POLARIZED LIGHT: (LINEARLY POLARIZED LIGHT):



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When the oscillation direction of electric field is confined to a single plane (⊥ to propagation direction) we call it linearly polarized light or plane polarized light. Linearly polarized light can be generated from unpolarized light using various methods such as selective absorption (dichroism), reflection, double refraction (birefringence) and scattering about which you will learn in upcoming sections.

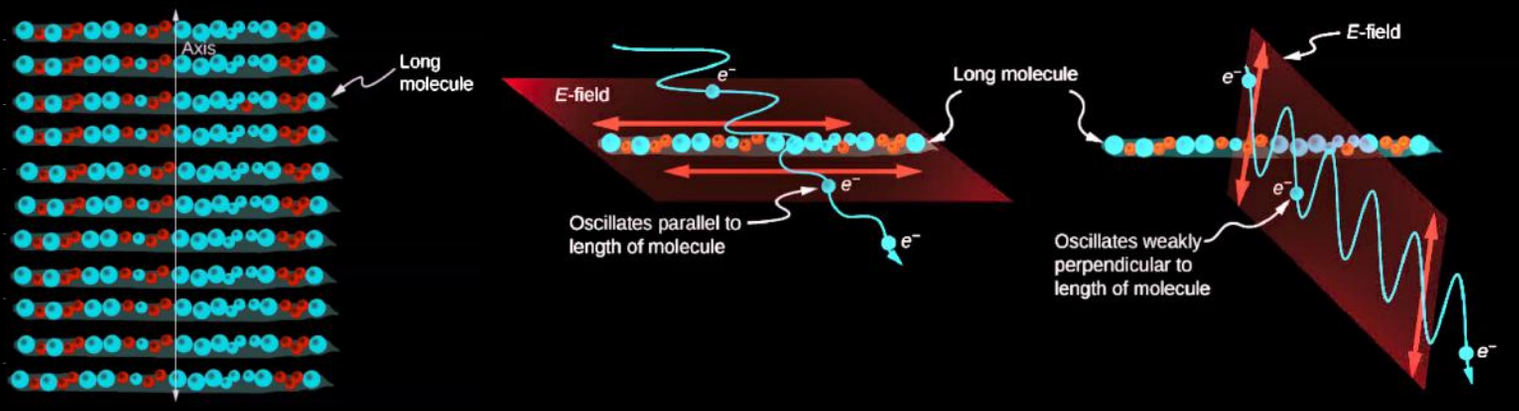
Plane Of Vibration And Plane Of Polarization:



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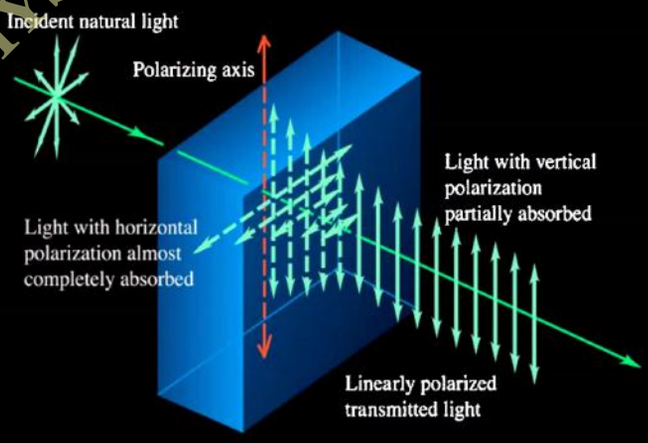
The plane containing oscillations of electric field is called plane of vibration and the plane perpendicular to it (magnetic field oscillation plane) is called plane of polarization (according Tamil Nadu Board Text book). [See figure above]

Polarization By Selective Absorption: (Dichroism)

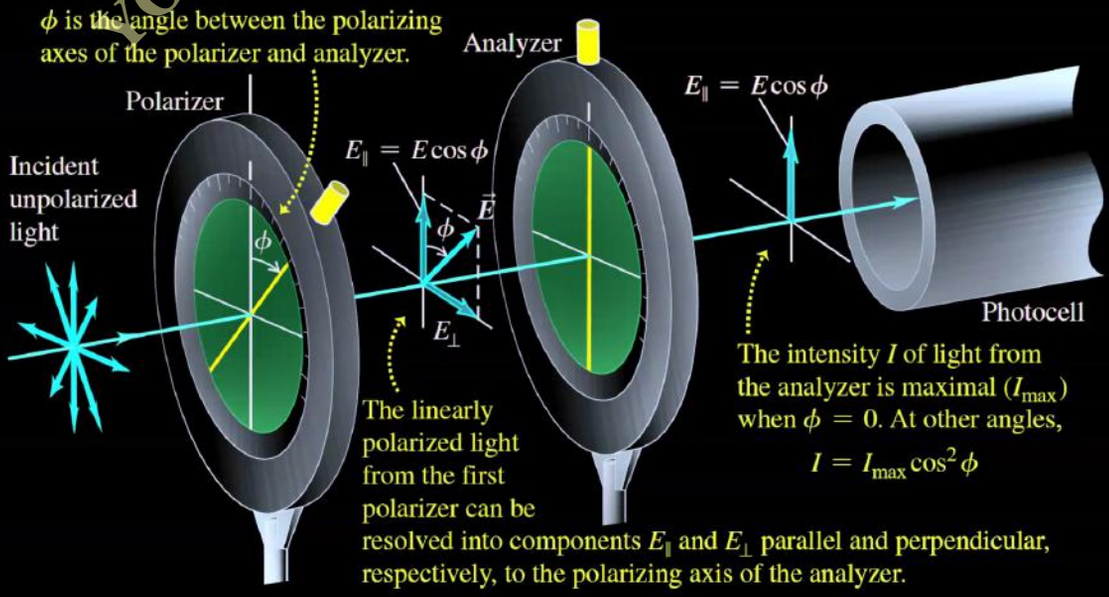


If long chain molecules in a material are aligned along a certain direction, then if EM wave is passed through them, the component of electric field along the chains gets absorbed (used in providing KE to the electrons in molecular orbitals) whereas the component ⊥ to the chains gets transmitted.

Naturally occurring crystal Tourmaline exhibits this property of Dichroism. Artificially such materials can be created in labs by stretching long chain polymer sheets. Polaroids (Trade name) are one such commonly available materials in form of sheets.



Polarizer Analyzer And Malus' Law:



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POLARIZER: Consider the setup for studying polarization as shown above. Here two polarizing sheets are arranged on which unpolarized light is incident from the left of first sheet. The first sheet on which the unpolarized light is incident is called polarizer.

TRANSMISSION AXIS: For polarizing sheets, the direction along which electric field oscillations are transmitted is called transmission axis

ANALYZER: The second polarizing sheet on which the light processed from the polarizer falls is called the analyzer. In a typical set up, the angle ϕ between transmission axes of polarizer and analyzer can be adjusted.

INTESITY OF LIGHT AFTER PROCESSING BY POLARIZER:

Electric field components in unpolarized light can be resolved in to components parallel and perpendicular to the transmission axes. The $\perp$ component is blocked and $\parallel$ component is transmitted thus only 50% of the unpolarized light is able to cross the polarizer. If I_u is intensity of unpolarized light and I_0, intensity after the polarizer then:

$$I_0 = I_u/2$$

INTENSITY AFTER PROCESSING BY ANALYZER:

Let angle between transmission axes of polarizer and analyzer be ϕ and let E be the electric field of polarized light after the polarizer, then the transmitted electric field from the analyzer must be $E_{\parallel} = E\cos\phi$. Now since intensity is proportional to square of electric field amplitude, accordingly, the intensity of beam transmitted by the analyzer is

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$I = I_0 \cos^2\phi$ ← MALUS' LAW

This formula for intensity is known as MALUS' law and it can be repeatedly applied for finding intensity through a stack of several polarizing sheets.

EXAMPLE: Unpolarized light falls on two stacked polarizing sheets such that transmitted intensity is 1/3 of incident light. Find the angle between transmission axes of sheets.

Solution:

$$I = I_0 \cos^2\phi = \frac{I_u}{2}\cos^2\phi = I_u/3 \Rightarrow \cos^2\phi = 2/3$$

$$\Rightarrow \phi = \cos^{-1}\left(\sqrt{2/3}\right) \text{ Ans.}$$

EXAMPLE: Unpolarized light of intensity 32 W/m^2 passes through three polarizers such that transmission axis of last polarizer is crossed with the first. If the intensity of emerging light is 3 W/m^2, what is the angle between transmission axes of first two polarizers? For what angle will the transmitted intensity be maximum?

Solution: Crossed axes means 90° angle between the axes of first and the third sheet. Let angle between axes of first and second be θ then by Malus' law, transmitted intensity:

$$I = \frac{I_u}{2}\cos^2\theta \sin^2\theta \quad (\because \text{from next sheet it will be } 90°-\theta).$$

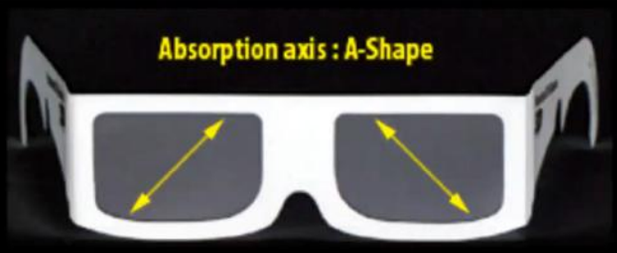
Thus $\frac{32}{2}\cos^2\theta \sin^2\theta = 3 \Rightarrow \sin^2 2\theta = 3/4 \Rightarrow \sin 2\theta = \sqrt{3}/2$

$\Rightarrow 2\theta = 60° \text{ or } 120° \Rightarrow \theta = 30° \text{ or } 60°$ Ans.

Now for maximum transmission we need to maximize $\sin^2\theta \times \cos^2\theta$ this happens for $\theta = 45°$ Ans.

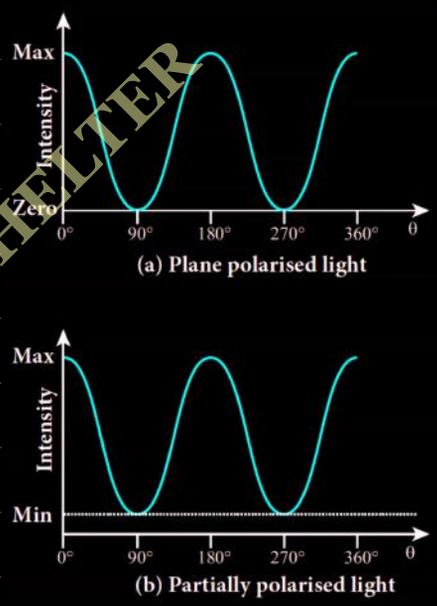
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→ Polarization by selective absorption is used in 3D glasses. Each side of the glasses transmits waves of different polarizations, thus the two eyes are fed with two slightly different images thus providing depth perception just like a real parallax.



Variation Of Intensity Through Dichroic Sheets:

If plane polarized light is incident on dichroic sheets, then intensity varies as $\cos^2\phi$ as the sheet is rotated where ϕ is angle between Electric field and transmission axis. In partially polarized light, there is some component of electric field in the perpendicular direction of major component. So even when polarizer axis is crossed with major component of electric field, some light will pass through.

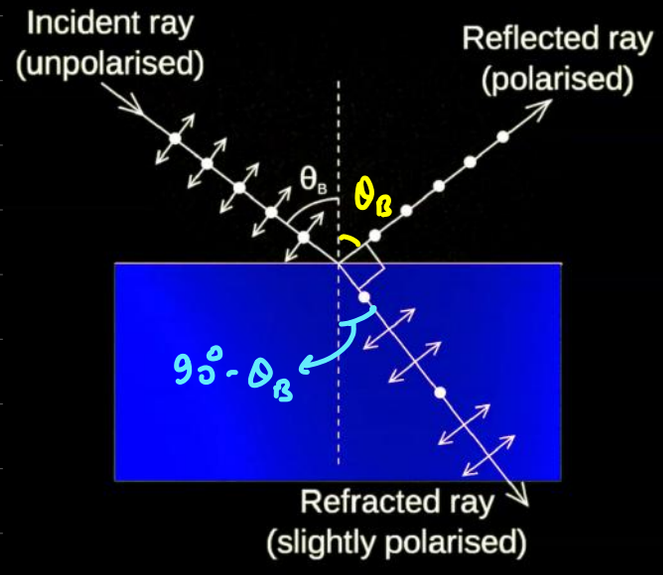


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Polarization By Reflection (Brewster's law):

When unpolarized light is incident on a refracting surface such that refracted beam and partially reflected beam are mutually perpendicular, the reflected light is completely plane polarized. The angle of incidence at which this happens is called Brewster Angle θ_B. This can be readily found from adjacent figure using Snell's law:

$$1 \sin\theta_B = \mu \cos\theta_B \quad \text{thus:}$$

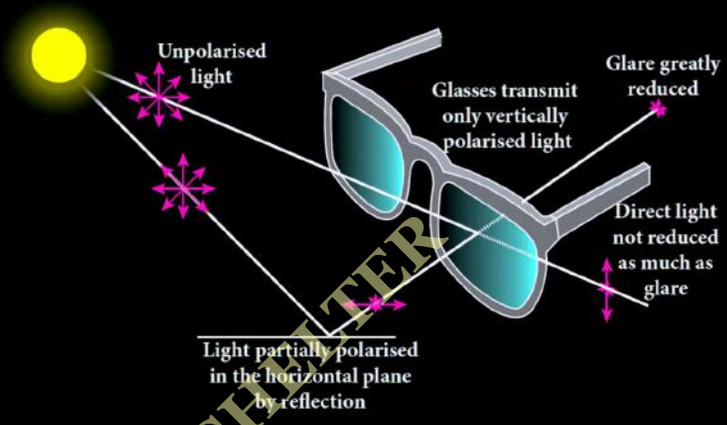


$$\tan\theta_B = \mu \quad \text{or} \quad \boxed{\theta_B = \tan^{-1}\mu}$$ ← Brewster Angle.

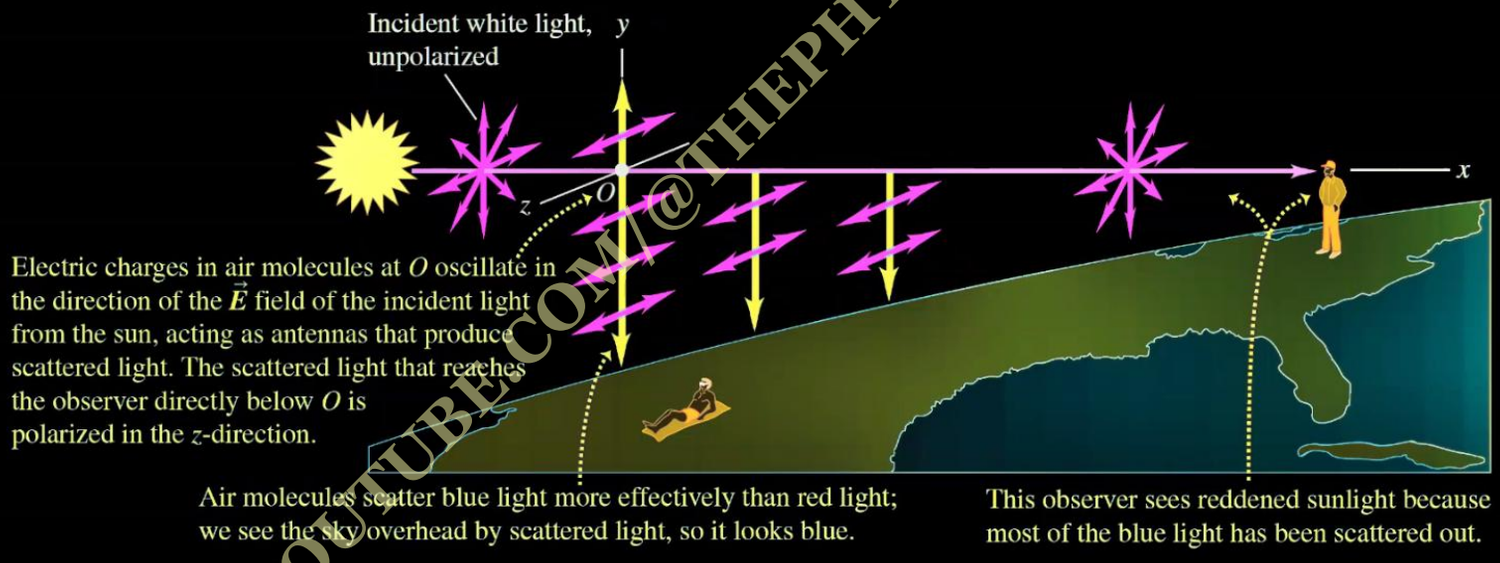
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Brewsten's law can be understood in terms of dipole oscillation radiation. Recall that when a dipole oscillates, it emits zero energy along the direction of oscillation. Thus at Brewster angle, electric field oscillations parallel to the reflected beam produce no radiation in the reflected beam.

→ Even at angles other than Brewster's angle, the reflected light is partly polarized and therefore polarizing glasses can be used to reduce glare of reflected light in seas and oceans. (Such as for navigators / Marine and shipping staff etc).



Colour Of Sky And Polarization By Scattering:

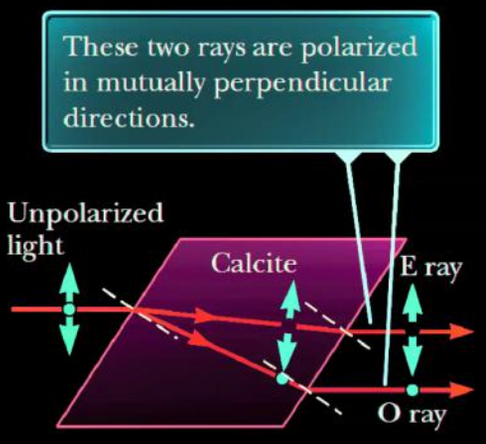


Consider a sun ray travelling horizontally as shown in the figure. Being unpolarized, the EM waves corresponding to this ray set the dipoles in atmospheric atoms in all directions. However when a dipole oscillates, it emits maximum energy in the direction ⊥ to the direction of oscillation. Thus to an observer looking vertically towards O, the energy recieved consists of horizontal oscillations and hence it is polarized light. Note that this observer is not looking at sunlight directly but he is looking at the scattered light from atmospheric atoms. The scattering of light is inversely related

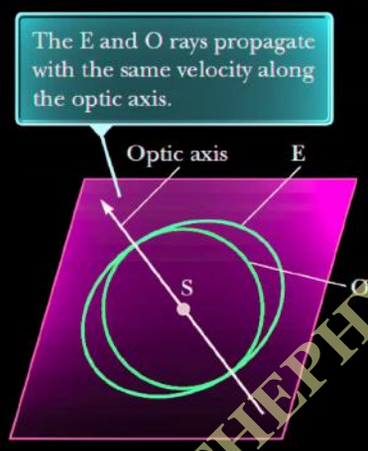
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to the fourth power of the wavelength. Thus, blue component of light scatters to the highest degree and therefore sky appears bluish to the observer looking vertically upwards. For an observer looking towards the horizon, the blue light appears to be missing because it has been scattered while traveling through the atmosphere. Therefore at the sun set or sunlight, sun appears reddish (Red photons are able to reach the observer while other colors are scattered to a comparatively larger degrees).

POLARIZATION BY DOUBLE REFRACTION (BIREFRINGENCE):



Unpolarized lightincident at an arbitrary direction on a calcite crystal splits into an ordinary (O) ray and an extraordi-nary (E) ray (not to scale).



A point source S inside a double-refracting crystal produces a spherical wave front corresponding to the ordinary (O) ray and an elliptical wave front corresponding to the extraordi-nary (E) ray.



A calcite crystal pro-duces a double image because it is a birefringent (double-refracting) material.

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In amorphous solids such as glass, speed of light is same in all direction however in certain crystalline solids such as quartz and calcite, speed is not same in all directions. In these materials speed of light depends on direction of propagation relative to planes of the crystal and plane of polarization. Such materials are characterized by two indices of refraction and hence called birefringent materials. (or Double refraction materials).

When unpolarized light enters a birefringent material, it may split in to **ordinary (O)** and **extraordinary (E)** rays. These two rays have mutually perpendicular polarizations and travel at different speeds in the material. These two speeds correspond to two indices

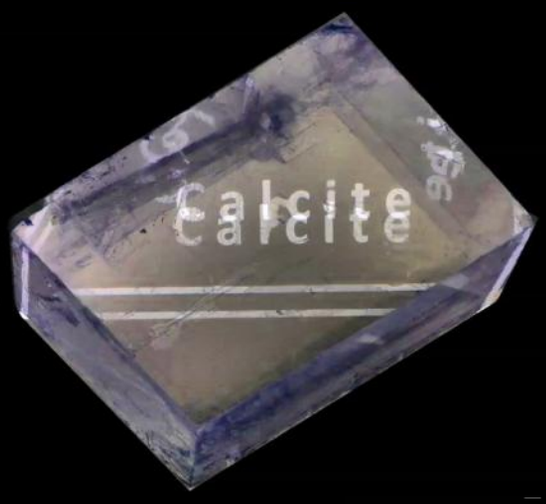
of refraction μ_o for ordinary ray and μ_E for the extraordinary ray.

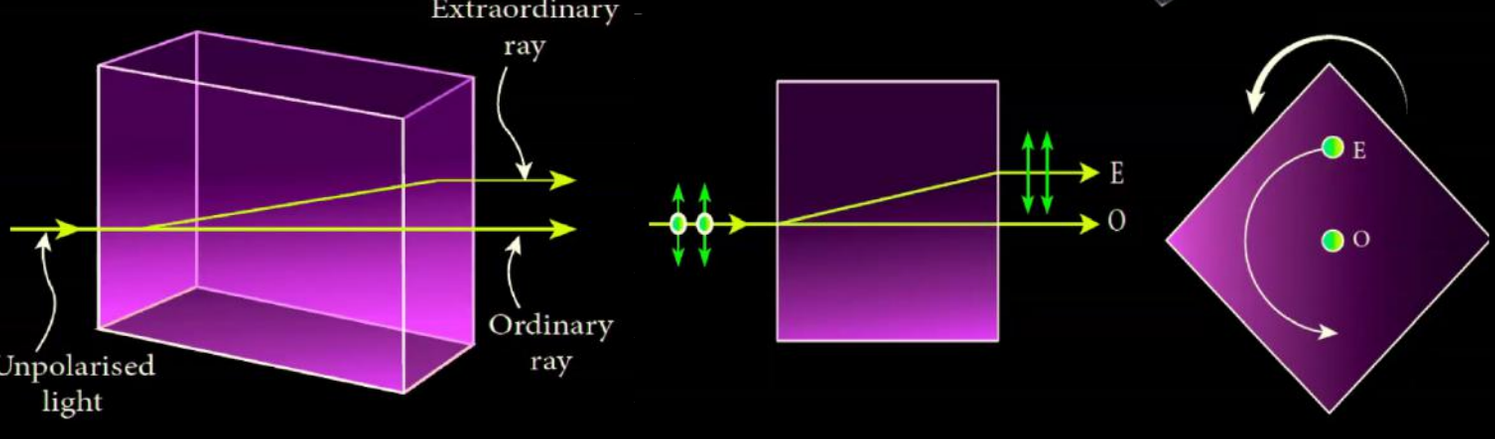
OPTIC AXIS: There is one particular direction in the crystal along which both ordinary and extraordinary rays have same speed. This direction is called optic axis. If unpolarized light enters the birefringent material from any other direction, the different indices of refraction cause the rays to split and travel in two different directions as shown in the first figure above.

The index of refraction μ_o is same for ordinary ray in all directions thus if one places a point source inside the crystal, the ordinary waves will spread out forming spherical wavefronts. However μ_E varies with direction of propagation and accordingly the wavefronts corresponding to extraordinary waves have wavefronts will elliptical cross sections. The difference in refractive indices is maximum in the direction perpendicular to the optic axis. For instance, for calcite, $\mu_o = 1.658$ ($\lambda = 589.3$ nm) and μ_E varies from 1.658 along optic axis to 1.486 perpendicular to the optic axis. (See the middle figure above).

IMAGE FORMATION FROM BIREFRINGENT MATERIALS:

Birefringent crystals generally produce two images due to different refractive indices. The two images are in different polarizations, thus if we see the images through a polaroid sheet, by rotating it at different angles, alternately the two images gradually appear and disappear.

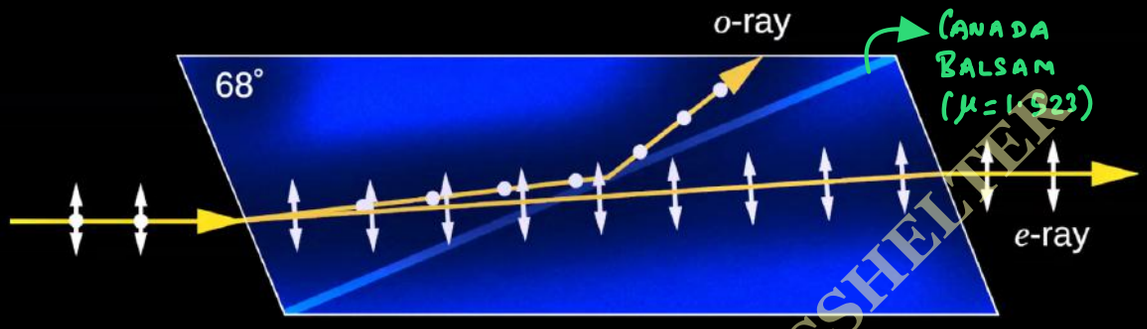




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Oridinary rays follow the usual laws of refraction whereas extraordinary rays do not follow the ordinary refraction laws. Thus when a calcite slab is placed on top of a dot on drawing sheet, we see two images of the dot and upon rotating the crystal, one image appears to rotate about the other.

NICOL PRISM:



In a Nicol prism, two calcite crystals (also known as iceland spar), are joined together with Canada Balsam ($\mu = 1.523$) as glue. The Nicol prism behaves as a denser medium for the ordinary ray as compared to the Canada Balsam (critical angle is 69°), thus ordinary ray undergoes TIR while for extraordinary rays, canada Balsam acts like slightly denser medium and thus extraordinary rays pass through the glue to the other crystal, thus we have plane polarized light generated out of unpolarized light.

→ Nicol prism can be used both as a polarizer and an analyzer.

DEGREE OF POLARIZATION: If partially polarized light is incident on a polaroid sheet, let I_{max} and I_{min} be the intensities of transmitted beams, then degree of polarization is defined as:

$$\eta = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

(optional).

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EXAMPLE: IRODOV: 5.159:

A beam of plane-polarized light falls on a polarizer which rotates about the axis of the ray with angular velocity $\omega = 21$ rad/s. Find the energy of light passing through the polarizer per one revolution if the flux of energy of the incident ray is equal to $\Phi_0 = 4.0$ mW.

Solution: Transmitted intensity at a general θ is $I_0 \cos^2\theta$ thus on an average, transmitted intensity is $I_0/2$. Thus energy passing in one revolution for power ϕ_0 : $U_1 = \frac{\phi_0}{2} \times \frac{2\pi}{\omega} = \frac{\phi_0 \pi}{\omega} = 0.6\,mJ$

EXAMPLE: IRODOV 5.160:

A beam of natural light falls on a system of $N = 6$ Nicol prisms whose transmission planes are turned each through an angle $\phi = 30°$ with respect to that of the foregoing prism. What fraction of luminous flux passes through this system?

Solution: First prism simply polarizes the unpolarized light while others reduce the intensity according to Malus' law. Thus passing fraction:

$$\tau = \frac{1}{2} \times (\cos^2\phi)^{N-1} = 0.12 \text{ Ans.}$$

EXAMPLE: Partially polarized light can be thought of as superposition of natural unpolarized light superposed with plane polarized light. If the degree of polarization of partially polarized light is $\eta = 0.25$ find the ratio of intensities of polarized and natural components in the beam.

Solution: Let natural component have an intensity I_1 and polarized component have an intensity I_2 then, $I_{max} = I_1/2 + I_2$ and $I_{min} = I_1/2$, Now $\eta = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{I_2}{I_1 + I_2}$

$$\Rightarrow \frac{I_2}{I_1} = \frac{\eta}{1-\eta} = \frac{1}{3} \text{ Ans.}$$

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EXAMPLE: IRODOV: 5.163:

A Nicol prism is placed in the way of partially polarized beam of light. When the prism is turned from the position of maximum transmission through an angle $\phi = 60°$, the intensity of transmitted light decreased by a factor of $\eta = 3.0$. Find the degree of polarization of incident light.

Solution: Let I_1 be natural component and I_2 be polarized component, then, $I_{max} = I_1/2 + I_2$ when prism is rotated through ϕ, intensity becomes $I = I_1/2 + I_2 \cos^2\phi$, thus according to the question:

$$\frac{(I_1/2 + I_2\cos^2\phi)}{I_1/2 + I_2} = \frac{1}{\eta} \quad (1), \quad \text{Also, } I_{max} = \frac{I_1}{2} + I_2, \quad I_{min} = I_1/2$$

Now degree of polarization: $f = \dfrac{I_{max} - I_{min}}{I_{max} + I_{min}} = \dfrac{I_2}{I_1 + I_2}$ (2)

From (1), $I_1/I_2 = \dfrac{2(1 - \eta\cos^2\phi)}{\eta - 1}$, Now from (2), $f = \dfrac{1}{1 + I_1/I_2}$

$$\Rightarrow f = \frac{\eta - 1}{1 - \eta\cos 2\phi} = 0.8 \text{ Ans.}$$

EXAMPLE: IRODOV 5.164:

Two identical imperfect polarizers are placed in the way of a natural beam of light. When the polarizers' planes are parallel, the system transmits $\eta = 10.0$ times more light than in the case of crossed planes. Find the degree of polarization of light produced

(a) by each polarizer separately

(b) by the whole system when the planes of the polarizers are parallel.

Solution: Let natural light comprise of two mutually $\perp$ polarized components each with intensity I_0. Further let α_1 be transmissivity for plane of oscillation along principal axis and let α_2 be transmissivity for component with oscillation $\perp$ to principal axis. Thus when the principal axes of two prisms are parallel, the transmission intensity is $I_{11} = \alpha_1^2 I_0 + \alpha_2^2 I_0$, Now with principal axes $\perp$ to each other the transmitted intensity $I_1 = \alpha_1\alpha_2 I_0 + \alpha_1\alpha_2 I_0$, Now according

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to the question: $\frac{I_\perp}{I_\parallel} = \frac{1}{\eta} \Rightarrow \frac{2\alpha_1\alpha_2}{\alpha_1^2+\alpha_2^2} = \frac{1}{\eta}$ ①

Now applying Componendo & Dividendo:

$$\frac{\alpha_1^2+\alpha_2^2-2\alpha_1\alpha_2}{\alpha_1^2+\alpha_2^2+2\alpha_1\alpha_2} = \frac{\eta-1}{\eta+1}$$

$$\Rightarrow \frac{\alpha_1-\alpha_2}{\alpha_1+\alpha_2} = \sqrt{\frac{\eta-1}{\eta+1}} \quad ②$$

($\alpha_2 < \alpha_1$ because in ideal case $\alpha_2 = 0$)

(a) Now Transmission for an ideal analyser at angle ϕ is $\alpha_1 I_0 \cos^2\phi + \alpha_2 I_0 \sin^2\phi$

$\Rightarrow I_t = \alpha_1 + (\alpha_2 - \alpha_1)\sin^2\phi$, this is maximum for $\sin^2\phi = 0$ and minimum for $\sin^2\phi = 1$ thus $I_{max} = \alpha_1 I_0$ and $I_{min} = \alpha_2 I_0$

Now degree of polarization: $f = \frac{I_{max}-I_{min}}{I_{max}+I_{min}} = \frac{\alpha_1-\alpha_2}{\alpha_1+\alpha_2} = \sqrt{\frac{\eta-1}{\eta+1}}$ Ans.

$= 0.905$ Ans.

(b) Now with two polarizers with axes parallel, the intensity through analyser (3rd polaroid), is $\alpha_1^2 I_0 \cos^2\phi + \alpha_2^2 I_0 \sin^2\phi$, this has max value of $\alpha_1^2 I_0$ and min value of $\alpha_2^2 I_0$, thus degree of polarization:

$$f = \frac{\alpha_1^2-\alpha_2^2}{\alpha_1^2+\alpha_2^2} = \frac{\alpha_1-\alpha_2}{\alpha_1+\alpha_2}\,\frac{(\alpha_1+\alpha_2)^2}{\alpha_1^2+\alpha_2^2} = \sqrt{\frac{\eta-1}{\eta+1}}\left(1+\frac{2\alpha_1\alpha_2}{\alpha_1^2+\alpha_2^2}\right)$$

$$\Rightarrow f = \sqrt{\frac{\eta-1}{\eta+1}}\left(1+\frac{1}{\eta}\right) \quad \text{(using ①)}$$

$$\Rightarrow f = \sqrt{1-\frac{1}{\eta^2}} = 0.995 \text{ ans.}$$

FRESNEL EQUATIONS: (OPTIONAL):

Fresnel equations for intensity of light reflected at the boundary between two dielectrics:

$$I_\perp' = I_\perp \frac{\sin^2(\theta_1-\theta_2)}{\sin^2(\theta_1+\theta_2)}, \quad I_\parallel' = I_\parallel \frac{\tan^2(\theta_1-\theta_2)}{\tan^2(\theta_1+\theta_2)}, \qquad \ldots(5.4d)$$

where $I_\perp$ and $I_\parallel$ are the intensities of incident light whose electric vector oscillations are respectively perpendicular and parallel to the plane of incidence.

θ_1: Incidence Angle

θ_2: Refraction Angle

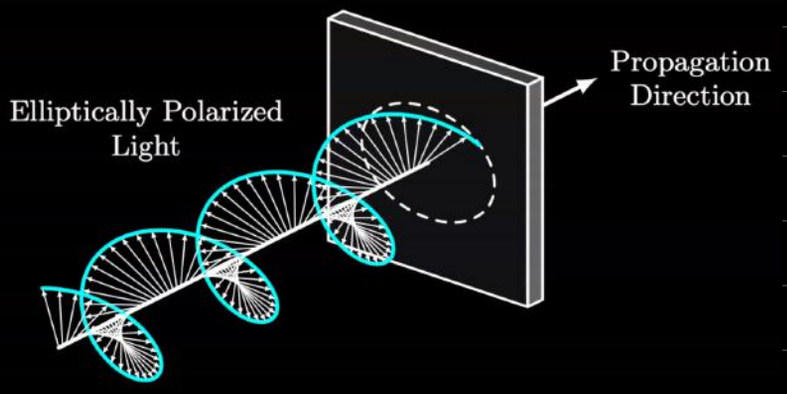
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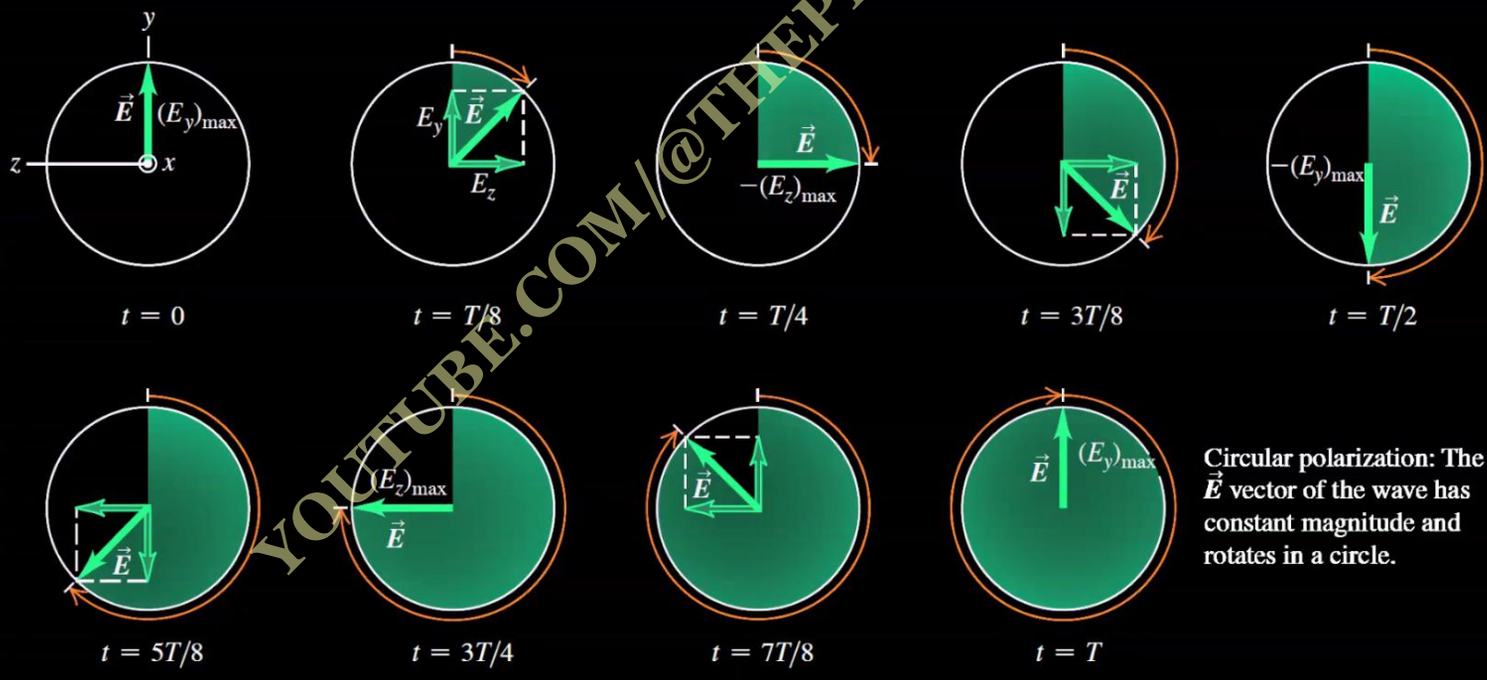
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Circular And Elliptical Polarization:

Recall from the chapter on SHM, when we superpose two mutually perpendicular SHMs with certain phase difference, the result is that the particle begins to trace an elliptical path in general whose eccentricity and sense of rotation depends on the phase difference.



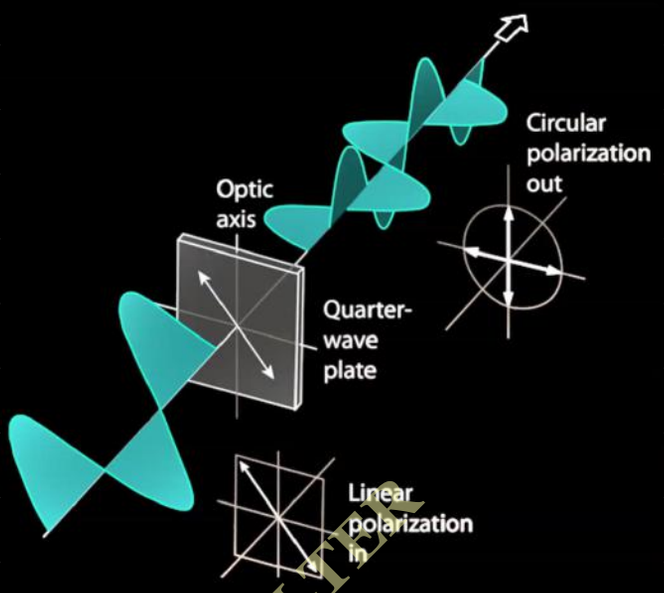
($x = A\cos\omega t$, $y = B\cos(\omega t + \phi)$). In a similar manner when two EM waves of mutually perpendicular polarizations are superposed, We get elliptically polarized light whose electric field vector traces an ellipse at any point in space. For equal amplitudes, and a phase difference of $\pi/2$, the electric field begins to trace a circle. Such a light is called circular polarized light.



If the wave is propagating towards you and the electric field appears to rotate clockwise, it is called right circularly polarized light and if it rotates anticlockwise it is called left circularly polarized light.

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Generating Circular Polarization From Plane Polarized Light:



To generate circular or elliptical polarized light, we have to generate a phase shift in two components of oscillation of electric field. This can be done using birefringent materials. For example if light is passed through a suitably oriented calcite crystal, The waves corresponding to two directions of polarization will travel with different speeds and hence will have some phase difference by the time they emerge out. Thus if input is a plane polarized light then if the phase difference of $\pi/2$ is introduced with equal amplitudes in mutually perpendicular directions, we will have circular polarized light. (Similarly if circular polarized light is passed through such a crystal, it will create a phase difference of another $\pi/2$ and hence generate plane polarized light from circular light.

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